

VAKIFBANK SOC — INTERNSHIP PRACTICE PROJECT

mlx-finetune-test

LoRA fine-tuning of a local LLM on Apple Silicon (MLX), taken end-to-end from a synthetic smoke test to a working phishing-email classifier served over a local API.

4B_{param}
Qwen3-4B-abliterated, 4-bit

18.6k_{emails}
Kaggle phishing dataset, raw

1000_{iters}
LoRA fine-tuning, batch 4

6/6_{correct}
held-out phishing/safe test

- 01

Validate the pipeline

DUMMY DATA

Before touching real data, trained the model on a synthetic task — prepend a fixed [SOC-TEST] marker to every reply — chosen so success is visible in the output, not just a loss number.

25 train / 5 valid rows

iters 200, batch 2

val loss 6.59 → 0.50

cross-lingual + paraphrase generalization held
- 02

Real data: phishing emails

KAGGLE

Swapped the synthetic task for a real one — classifying emails as Phishing or Safe. Raw data needed real cleanup before it was trainable.

1,068 duplicates dropped (leakage risk)

length capped at 1,000 chars (tail ran to 17M)

9,356 of 18,650 emails kept

stratified 90/10 split → 8,420 train / 936 valid

03 Fine-tune — and don't trust val loss alone THE KEY LESSON

Val loss looked best early, but the low-loss checkpoint had only learned to guess the majority class. Actual generation output was the only reliable signal.

iters 1000, batch 4

8,420 train rows after cleanup

Iter 200 had the lowest early val loss (3.03) — and predicted "Safe" for every email, including obvious phishing. It had learned the class balance, not the task. Training continued to iter 1000, and checkpoints were verified against 6 hand-written emails before picking one.

val loss — iter 1: 4.76 → 200: 3.03 → 400: 3.22 → 700: 2.96 → 1000: 3.15 (non-monotonic)

04 Serve it LLAMA.CPP

The finished adapter is fused into the base model, converted to GGUF, quantized, and served behind an OpenAI-compatible API — with auth, not left open.

fuse → GGUF → Q4_K_M quantize (8GB → 2.4GB)

llama-server, OpenAI-compatible endpoint

API key required, CORS restricted to localhost

IN SHORT

Built and verified a full local fine-tuning pipeline on Apple Silicon: took a 4B-parameter open model, taught it a task with LoRA, and served the result behind an authenticated API — first proving the mechanics with a synthetic task, then repeating it on a real one (phishing email detection).

The main technical finding: **the validation-loss curve was not a reliable stopping signal** — the checkpoint with the lowest loss had actually learned to guess the majority class, not the task. Catching this required checking real model output by hand, not just the metric.